



National  
Qualifications  
2023

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# **2023 Mathematics**

## **Higher - Paper 2**

### **Finalised Marking Instructions**

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## General marking principles for Higher Mathematics

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

*For each question, the marking instructions are generally in two sections:*

- *generic scheme – this indicates why each mark is awarded*
- *illustrative scheme – this covers methods which are commonly seen throughout the marking*

*In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.*

- Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.
- One mark is available for each •. There are no half marks.
- If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.
- Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.
- Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.
- If an error is trivial, casual or insignificant, for example  $6 \times 6 = 12$ , candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.
- If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 1 \text{ or } 3$$

(i) **Horizontal/vertical marking**

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

$$\begin{array}{cc} \bullet^5 & \bullet^6 \\ \bullet^5 & x = 2 \quad x = -4 \\ \bullet^6 & y = 5 \quad y = -7 \end{array}$$

Horizontal:  $\bullet^5 x = 2$  and  $x = -4$     Vertical:  $\bullet^5 x = 2$  and  $y = 5$   
 $\bullet^6 y = 5$  and  $y = -7$      $\bullet^6 x = -4$  and  $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$$\frac{15}{12} \text{ must be simplified to } \frac{5}{4} \text{ or } 1\frac{1}{4} \quad \frac{43}{1} \text{ must be simplified to } 43$$

$$\frac{15}{0.3} \text{ must be simplified to } 50 \quad \frac{4\cancel{5}}{3} \text{ must be simplified to } \frac{4}{15}$$

$$\sqrt{64} \text{ must be simplified to } 8^*$$

\*The square root of perfect squares up to and including 144 must be known.

(k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

(l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example  
 $(x^3 + 2x^2 + 3x + 2)(2x + 1)$  written as  
 $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$   
 $= 2x^4 + 5x^3 + 8x^2 + 7x + 2$  gains full credit
- repeated error within a question, but not between questions or papers

(m) In any 'Show that...' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

- (n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.
- (o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.
- (p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

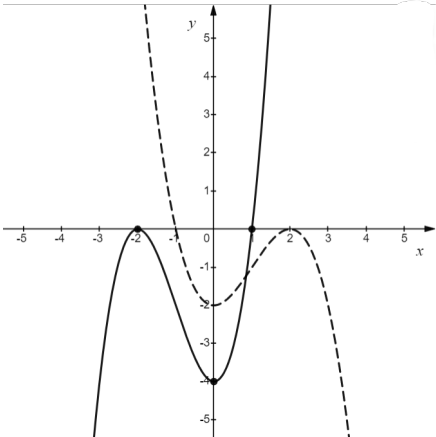
|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks.                             | Strategy 2 attempt 1 is worth 1 mark.                              |
| Strategy 1 attempt 2 is worth 4 marks.                             | Strategy 2 attempt 2 is worth 5 marks.                             |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |  | Generic scheme                                                                                                                                                                                       | Illustrative scheme                                                                                                                                                                                          | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> find gradient of QR</li> <li>•<sup>2</sup> use property of perpendicular lines</li> <li>•<sup>3</sup> determine equation of altitude</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>-\frac{1}{3}</math> or <math>-\frac{5}{15}</math></li> <li>•<sup>2</sup> 3</li> <li>•<sup>3</sup> <math>y = 3x - 16</math></li> </ul>             | 3        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| 1. • <sup>3</sup> is only available to candidates who find and use a perpendicular gradient.<br>2. The gradient of the perpendicular bisector must appear in fully simplified form at • <sup>2</sup> or • <sup>3</sup> stage for • <sup>3</sup> to be awarded - see Candidate B.<br>3. • <sup>3</sup> is not available as a consequence of using the midpoint of QR and the point P.<br>4. At • <sup>3</sup> , accept any arrangement of a candidate's equation where constant terms have been simplified. |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| Candidate A - BEWARE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  | Candidate B - unsimplified gradient                                                                                                                                                                  |                                                                                                                                                                                                              |          |
| Correct equation from incorrect substitution                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| $m = \frac{13 - (-2)}{3 - (8)} = 3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |  | $m = -\frac{5}{15}$                                                                                                                                                                                  |                                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | $m_{\perp} = \frac{15}{5}$                                                                                                                                                                           |                                                                                                                                                                                                              |          |
| $y = 3x - 16$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  | $15x - 5y - 80 = 0$                                                                                                                                                                                  |                                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> determine gradient of the line</li> <li>•<sup>5</sup> use <math>m = \tan \theta</math> to find the angle</li> </ul>                             | <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>m = \frac{1}{2}</math> or <math>\tan \theta = \frac{1}{2}</math></li> <li>•<sup>5</sup> <math>26.6^{\circ}</math> or 0.4636... radians</li> </ul> | 2        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| 5. Do not penalise the omission of units at • <sup>5</sup> .<br>6. Accept any answers which round to $27^{\circ}$ or 0.46 radians.<br>7. For $27^{\circ}$ or 0.46 radians without working award 2/2.<br>8. Where candidates find the angle of the altitude or other sides with the positive direction of the $x$ -axis only • <sup>5</sup> is available.                                                                                                                                                   |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| Candidate C - no reference to tan                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |  | Candidate D - BEWARE                                                                                                                                                                                 |                                                                                                                                                                                                              |          |
| $m = \frac{4}{8}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |  | $m = \frac{1}{2}$                                                                                                                                                                                    |                                                                                                                                                                                                              |          |
| $26.6^{\circ}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | $\theta = \tan^{-1} \frac{1}{2}$                                                                                                                                                                     |                                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | $\theta = 26.6^{\circ}$                                                                                                                                                                              |                                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | Stating tan rather than $\tan^{-1}$                                                                                                                                                                  |                                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | See general marking principle (g)                                                                                                                                                                    |                                                                                                                                                                                                              |          |
| Candidate E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |
| $\tan^{-1}(3) = 72^{\circ}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                                                      |                                                                                                                                                                                                              |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | Generic scheme                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                      | Max mark                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> calculate y-coordinate</li> <li>•<sup>2</sup> differentiate</li> <li>•<sup>3</sup> calculate the gradient</li> <li>•<sup>4</sup> find equation of line</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> -1</li> <li>•<sup>2</sup> <math>10x^4 - 3</math></li> <li>•<sup>3</sup> 7</li> <li>•<sup>4</sup> <math>y = 7x - 8</math></li> </ul> | 4                                                                                                                                                                                                            |
| <b>Notes:</b> <ol style="list-style-type: none"> <li>Only •<sup>1</sup> is available to candidates who integrate.</li> <li>•<sup>4</sup> is only available where candidates attempt to find the gradient by substituting into their derivative.</li> <li>The appearance of <math>10x^4 - 3</math> gains •<sup>2</sup>.</li> <li>•<sup>3</sup> is not available for <math>y = 7</math>. However, where 7 is then used as the gradient of the straight line, •<sup>3</sup> may be awarded - see Candidates B, C &amp; D.</li> <li>•<sup>4</sup> is not available as a consequence of using a perpendicular gradient.</li> </ol> |  |  |                                                                                                                                                                                                                        |                                                                                                                                                                                          |                                                                                                                                                                                                              |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                        |                                                                                                                                                                                          |                                                                                                                                                                                                              |
| <b>Candidate A</b><br>$\frac{dy}{dx} = 10x^4 - 3$<br>$y = 7$<br>$m = -3$<br>$y = -3x + 10$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  | <ul style="list-style-type: none"> <li>•<sup>2</sup> ✓</li> <li>•<sup>1</sup> ✗</li> <li>•<sup>3</sup> ✗</li> <li>•<sup>4</sup> <span style="border: 1px solid red; padding: 2px;">✓<sub>2</sub></span></li> </ul>     | <b>Candidate B - incorrect notation</b><br>$y = -1$<br>$y = 10x^4 - 3$<br>$y = 7$<br>$y + 1 = 7(x - 1)$<br>$y = 7x - 8$                                                                  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓ - BoD</li> <li>•<sup>2</sup> ✓</li> <li>•<sup>3</sup> ✓ - BoD</li> <li>•<sup>4</sup> ✓</li> </ul>                                                     |
| <b>Candidate C - use of values in equation</b><br>$y = -1$<br>$\frac{dy}{dx} = 10x^4 - 3$<br>$\frac{dy}{dx} = 7$<br>$y = 7$<br>$y + 1 = 7(x - 1)$<br>$y = 7x - 8$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓ - BoD</li> <li>•<sup>2</sup> ✓</li> <li>•<sup>3</sup> ✓</li> <li>•<sup>4</sup> ✓</li> </ul>                                                                     | <b>Candidate D - incorrect notation</b><br>$y = -1$<br>$\frac{dy}{dx} = 10x^4 - 3$<br>$y = 7$                                                                                            | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓ - BoD</li> <li>•<sup>2</sup> ✓</li> <li>•<sup>3</sup> ✗</li> </ul> <p>Evidence for •<sup>3</sup> would need to appear in the equation of the line</p> |
| <b>Candidate E</b><br>$y = -1$<br>$\frac{dy}{dx} = 10x^4 - 3 = 0$<br>$10(1)^4 - 3 = 0$<br>$m = 7$<br>$y = 7x - 8$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✓</li> <li>•<sup>3</sup> ✗</li> <li>•<sup>4</sup> <span style="border: 1px solid red; padding: 2px;">✓<sub>1</sub></span></li> </ul>     |                                                                                                                                                                                          |                                                                                                                                                                                                              |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | Generic scheme                                                                                            | Illustrative scheme                                                                                                                          | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  | <p>•<sup>1</sup> start to integrate</p> <p>•<sup>2</sup> complete integration</p>                         | <p>•<sup>1</sup> <math>7 \sin\left(4x + \frac{\pi}{3}\right) \dots</math></p> <p>•<sup>2</sup> <math>\dots \times \frac{1}{4} + c</math></p> | 2        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                                                                                                           |                                                                                                                                              |          |
| <p>1. Award •<sup>1</sup> for any appearance of <math>(+)7 \sin\left(4x + \frac{\pi}{3}\right)</math> regardless of any constant multiplier.</p> <p>2. Candidates who work in degrees from the start cannot gain •<sup>1</sup>, however •<sup>2</sup> is still available - see Candidate C.</p> <p>3. Where candidates use any other invalid approach, eg <math>7 \sin\left(4x + \frac{\pi}{3}\right)^2</math>, <math>\int \left(7 \cos 4x + \cos \frac{\pi}{3}\right) dx</math> or <math>7 \sin 4x + \frac{\pi}{3}</math> award 0/2. However, see Candidate E.</p> <p>4. Do not penalise the appearance of an integral sign and/or <math>dx</math> throughout.</p> |  |  |                                                                                                           |                                                                                                                                              |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |                                                                                                           |                                                                                                                                              |          |
| <b>Candidate A - using addition formula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  | <b>Candidate B</b>                                                                                        |                                                                                                                                              |          |
| $\int \left(7 \cos 4x \cos \frac{\pi}{3} - 7 \sin 4x \sin \frac{\pi}{3}\right) dx$ $= \frac{7}{4} \sin 4x \cos \frac{\pi}{3} + \frac{7}{4} \cos 4x \sin \frac{\pi}{3} \dots$ $= \frac{7}{4} \sin 4x \left(\frac{1}{2}\right) + \frac{7}{4} \cos 4x \left(\frac{\sqrt{3}}{2}\right) + c$                                                                                                                                                                                                                                                                                                                                                                             |  |  | $\frac{7}{4} \sin\left(4x + \frac{\pi}{3}\right)$ $= \frac{7}{4} \sin\left(4x + \frac{\pi}{3}\right) + c$ |                                                                                                                                              |          |
| • <sup>1</sup> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  | • <sup>1</sup> ✓                                                                                          |                                                                                                                                              |          |
| • <sup>2</sup> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  | • <sup>2</sup> ✓                                                                                          |                                                                                                                                              |          |
| <b>Candidate C - working in degrees</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | <b>Candidate D - integrating over two lines</b>                                                           |                                                                                                                                              |          |
| $\int 7 \cos(4x + 60) dx$ $= 7 \sin(4x + 60) \times \frac{1}{4} + c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | $7 \sin\left(4x + \frac{\pi}{3}\right)$ $= \frac{7}{4} \sin\left(4x + \frac{\pi}{3}\right) + c$           |                                                                                                                                              |          |
| • <sup>1</sup> ✗ • <sup>2</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | • <sup>1</sup> ✓                                                                                          |                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  | • <sup>2</sup> ✗                                                                                          |                                                                                                                                              |          |
| <b>Candidate E - integrating in part</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | <b>Candidate F - insufficient evidence of integration</b>                                                 |                                                                                                                                              |          |
| $-\frac{7}{4} \sin\left(4x + \frac{\pi}{3}\right) + c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  | $\frac{7}{4} \cos\left(4x + \frac{\pi}{3}\right) + c$                                                     |                                                                                                                                              |          |
| • <sup>1</sup> ✗ • <sup>2</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | • <sup>1</sup> ✗ • <sup>2</sup> ✗                                                                         |                                                                                                                                              |          |

| Question |  |  | Generic scheme                                                                                                                                               | Illustrative scheme                                                                                                                                                                                                                                             | Max mark |
|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> reflect in the <math>y</math>-axis</li> <li>•<sup>2</sup> apply appropriate vertical scaling</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> cubic graph with max at <math>(-2, 0)</math> and passing through <math>(1, 0)</math></li> <li>•<sup>2</sup>  </li> </ul> | 2        |

**Notes:**

1. Where candidates do not sketch a cubic function award 0/2.
2. For transformations of the form  $f(-x) + k$  or  $-f(x + k)$  award 0/2.
3. If the transformation has not been applied to all coordinates, award 0/2.

| Question                       | Generic scheme                         | Illustrative scheme        | Max mark |       |
|--------------------------------|----------------------------------------|----------------------------|----------|-------|
| 4.                             | (continued)                            |                            |          |       |
| Commonly Observed Responses:   |                                        |                            |          |       |
| Function                       | Transformation of $(-1,0)$ and $(2,0)$ | Transformation of $(0,-2)$ | Shape    | Award |
| Incorrect orientation          | $(-2,0)$ and $(1,0)$                   | $(0,-4)$                   |          | 0/2   |
| $-2f(x)$                       | $(-1,0)$ and $(2,0)$                   | $(0,4)$                    |          | 1/2   |
| $-2f(-x)$                      | $(-2,0)$ and $(1,0)$                   | $(0,4)$                    |          | 1/2   |
| $-2f(-2x)$                     | $(-1,0)$ and $(\frac{1}{2},0)$         | $(0,4)$                    |          | 0/2   |
| $-2f\left(-\frac{x}{2}\right)$ | $(-4,0)$ and $(2,0)$                   | $(0,4)$                    |          | 0/2   |
| $2f(x)$                        | $(-1,0)$ and $(2,0)$                   | $(0,-4)$                   |          | 1/2   |
| $2f(2x)$                       | $(-\frac{1}{2},0)$ and $(1,0)$         | $(0,-4)$                   |          | 1/2   |
| $2f\left(\frac{x}{2}\right)$   | $(-2,0)$ and $(4,0)$                   | $(0,-4)$                   |          | 1/2   |
| $2f\left(-\frac{x}{2}\right)$  | $(-4,0)$ and $(2,0)$                   | $(0,-4)$                   |          | 1/2   |
| $2f(x-1)$                      | $(0,0)$ and $(3,0)$                    | $(1,-4)$                   |          | 1/2   |
| $f(-x)$                        | $(-2,0)$ and $(1,0)$                   | $(0,-2)$                   |          | 1/2   |
| $\frac{1}{2}f(-x)$             | $(-2,0)$ and $(1,0)$                   | $(0,-1)$                   |          | 1/2   |
| $f(2x)$                        | $(-\frac{1}{2},0)$ and $(1,0)$         | $(0,-2)$                   |          | 0/2   |
| $f(-2x)$                       | $(-1,0)$ and $(\frac{1}{2},0)$         | $(0,-2)$                   |          | 0/2   |
| $f\left(-\frac{x}{2}\right)$   | $(-4,0)$ and $(2,0)$                   | $(0,-2)$                   |          | 0/2   |
| $-f\left(\frac{x}{2}\right)$   | $(-2,0)$ and $(4,0)$                   | $(0,2)$                    |          | 0/2   |
| $-f\left(-\frac{x}{2}\right)$  | $(-4,0)$ and $(2,0)$                   | $(0,2)$                    |          | 0/2   |

| Question                                                                                                                                                                                                                                                                                 |  |  | Generic scheme                                                                                                                                                                         | Illustrative scheme                                                                                                                                                            | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5.                                                                                                                                                                                                                                                                                       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> start to differentiate</li> <li>•<sup>2</sup> complete differentiation</li> <li>•<sup>3</sup> calculate rate of change</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>4(3-2x)^3 \dots</math></li> <li>•<sup>2</sup> <math>\dots \times (-2)</math></li> <li>•<sup>3</sup> 1000</li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                        |                                                                                                                                                                                |          |
| 1. Correct answer with no working, award 0/3.<br>2. Accept $4u^3 \times (-2)$ where $u = 3 - 2x$ for • <sup>1</sup> .<br>3. Where candidates evaluate $f(4)$ , award 0/3, see Candidate B.<br>4. • <sup>3</sup> is only available for evaluating expressions equivalent to $k(3-2x)^3$ . |  |  |                                                                                                                                                                                        |                                                                                                                                                                                |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                      |  |  |                                                                                                                                                                                        |                                                                                                                                                                                |          |
| <b>Candidate A</b>                                                                                                                                                                                                                                                                       |  |  | <b>Candidate B - evaluating f(x)</b>                                                                                                                                                   |                                                                                                                                                                                |          |
| $f'(x) = 4(3-2x)^3 \times (-2)$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$f'(x) = 8(3-2x)^3$<br>$f'(4) = -1000$ • <sup>3</sup> ✗                                                                                                                                                             |  |  | $f'(x) = (3-2x)^4$ • <sup>1</sup> ✗ • <sup>2</sup> ✗<br>$f'(4) = 625$ • <sup>3</sup> ✗                                                                                                 |                                                                                                                                                                                |          |
| <b>Candidate C - differentiating over two lines</b>                                                                                                                                                                                                                                      |  |  | <b>Candidate D - differentiating over two lines</b>                                                                                                                                    |                                                                                                                                                                                |          |
| $4(3-2x)^3$ • <sup>1</sup> ✓<br>$4(3-2x)^3 \times 2$ • <sup>2</sup> ✗<br>$-1000$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                           |  |  | $4(3-2x)^3$ • <sup>1</sup> ✓<br>$4(3-2x)^3 \times -2$ • <sup>2</sup> ^<br>$1000$ • <sup>3</sup> ✓ <sub>1</sub>                                                                         |                                                                                                                                                                                |          |
| <b>Candidate E - insufficient evidence for mark 1</b>                                                                                                                                                                                                                                    |  |  | <b>Candidate F</b>                                                                                                                                                                     |                                                                                                                                                                                |          |
| $f'(x) = 8(3-2x)^3$ • <sup>1</sup> ✗ • <sup>2</sup> ✗<br>$f'(4) = -1000$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                   |  |  | $4(3-2x)^3$ • <sup>1</sup> ✓ • <sup>2</sup> ^<br>$-500$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                  |                                                                                                                                                                                |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                             | Max mark |
|----------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.       |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> equate composite function to <math>x</math></li> <li>•<sup>2</sup> write <math>f(f^{-1}(x))</math> in terms of <math>f^{-1}(x)</math></li> <li>•<sup>3</sup> state inverse function</li> </ul> | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>f(f^{-1}(x)) = x</math></li> <li>•<sup>2</sup> <math>x = \frac{2}{f^{-1}(x)} + 3</math></li> <li>•<sup>3</sup> <math>f^{-1}(x) = \frac{2}{x-3}</math></li> </ul>                                                                                                     | 3        |
|          |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> write as <math>y = f(x)</math> and start to rearrange</li> <li>•<sup>2</sup> express <math>x</math> in terms of <math>y</math></li> <li>•<sup>3</sup> state inverse function</li> </ul>        | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = f(x) \Rightarrow x = f^{-1}(y)</math><br/><math>y - 3 = \frac{2}{x}</math></li> <li>•<sup>2</sup> <math>x = \frac{2}{y-3}</math></li> <li>•<sup>3</sup> <math>f^{-1}(y) = \frac{2}{y-3}</math><br/><math>\Rightarrow f^{-1}(x) = \frac{2}{x-3}</math></li> </ul> |          |

#### Notes:

1. In Method, 1 accept  $x = \frac{2}{f^{-1}(x)} + 3$  for •<sup>1</sup> and •<sup>2</sup>.
2. In Method 2, accept ' $y - 3 = \frac{2}{x}$ ', without reference to  $y = f(x) \Rightarrow x = f^{-1}(y)$  at •<sup>1</sup>.
3. In Method 2, accept  $f^{-1}(x) = \frac{2}{x-3}$  without reference to  $f^{-1}(y)$  at •<sup>3</sup>.
4. In Method 2, beware of candidates with working where each line is not mathematically equivalent - see Candidates A and B for example.
5. At •<sup>3</sup> stage, accept  $f^{-1}$  written in terms of any dummy variable eg  $f^{-1}(y) = \frac{2}{y-3}$ .
6.  $y = \frac{2}{x-3}$  does not gain •<sup>3</sup>.
7.  $f^{-1}(x) = \frac{2}{x-3}$  with no working gains 3/3.
8. In Method 2, where candidates make multiple algebraic errors at the •<sup>2</sup> stage, •<sup>3</sup> is still available.

| Question                                                                                                                                                                            | Generic scheme                                                                                                   | Illustrative scheme                                                                                                                                                                                                                              | Max mark                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 6. (continued)                                                                                                                                                                      |                                                                                                                  |                                                                                                                                                                                                                                                  |                                                                                            |
| Commonly Observed Responses:                                                                                                                                                        |                                                                                                                  |                                                                                                                                                                                                                                                  |                                                                                            |
| <b>Candidate A</b><br>$f(x) = \frac{2}{x} + 3$<br>$y = \frac{2}{x} + 3$<br>$y - 3 = \frac{2}{x}$<br>$x = \frac{2}{y - 3}$<br>$y = \frac{2}{x - 3}$<br>$f^{-1}(x) = \frac{2}{x - 3}$ | $\bullet^1 \checkmark \bullet^2 \checkmark$<br>$\bullet^3 \times$                                                | <b>Candidate B</b><br>$f(x) = \frac{2}{x} + 3$<br>$y = \frac{2}{x} + 3$<br>$x = \frac{2}{y} + 3$<br>$x - 3 = \frac{2}{y}$<br>$y = \frac{2}{x - 3}$<br>$f^{-1}(x) = \frac{2}{x - 3}$                                                              | $\bullet^1 \times$<br>$\bullet^2 \boxed{\checkmark}_1$<br>$\bullet^3 \boxed{\checkmark}_1$ |
| <b>Candidate C - BEWARE</b><br>$f'(x) = \dots$                                                                                                                                      | $\bullet^3 \times$                                                                                               | <b>Candidate D</b><br>$x \rightarrow \frac{1}{x} \rightarrow \frac{2}{x} \rightarrow \frac{2}{x} + 3 = f(x)$<br>$\times 2 \rightarrow + 3$<br>$\therefore - 3 \rightarrow \div 2$<br>$\frac{2}{x - 3}$ (invert)<br>$f^{-1}(x) = \frac{2}{x - 3}$ | $\bullet^1 \checkmark$<br>$\bullet^2 \checkmark$<br>$\bullet^3 \checkmark$                 |
| <b>Candidate E</b><br>$\therefore$<br>$f^{-1}(x) = \left(\frac{x - 3}{2}\right)^{-1}$                                                                                               | $\bullet^1 \checkmark \bullet^2 \checkmark$<br>$\bullet^3 \checkmark$                                            | <b>Candidate F</b><br>$\therefore$<br>$f^{-1}(x) = \sqrt[1]{\frac{x - 3}{2}}$                                                                                                                                                                    | $\bullet^1 \checkmark \bullet^2 \checkmark$<br>$\bullet^3 \checkmark$                      |
| <b>Candidate G</b><br>$y = \frac{2}{x} + 3$<br>$xy = 5$<br>$x = \frac{5}{y}$<br>$f^{-1}(x) = \frac{5}{x}$<br><b>However</b><br>$f^{-1}(x) = \frac{2 + 3}{x}$                        | $\bullet^1 \times$<br>$\bullet^2 \boxed{\checkmark}_2$<br>$\bullet^3 \boxed{\checkmark}_1$<br>$\bullet^3 \wedge$ |                                                                                                                                                                                                                                                  |                                                                                            |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark |
|----------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use double angle formula to express equation in terms of <math>\sin x^\circ</math></li> <li>•<sup>2</sup> arrange in standard quadratic form</li> <li>•<sup>3</sup> factorise or use quadratic formula</li> <li>•<sup>4</sup> solve for <math>\sin x^\circ</math></li> <li>•<sup>5</sup> solve for <math>x</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\dots = 3(1 - 2\sin^2 x^\circ)</math></li> <li>•<sup>2</sup> <math>6\sin^2 x^\circ + \sin x^\circ - 1 = 0</math></li> <li>•<sup>3</sup> <math>(3\sin x^\circ - 1)(2\sin x^\circ + 1) (= 0)</math><br/>or <math>\sin x^\circ = \frac{-1 \pm \sqrt{25}}{12}</math></li> <li>•<sup>4</sup> <math>\sin x^\circ = \frac{1}{3},</math>      •<sup>5</sup> <math>\sin x^\circ = -\frac{1}{2}</math></li> <li>•<sup>5</sup> 19.47 ..., 160.52 ..., 210, 330</li> </ul> | 5        |

#### Notes:

1. Substituting  $1 - 2\sin^2 A$  or  $1 - 2\sin^2 \alpha$  for  $\cos 2x^\circ$  at the •<sup>1</sup> stage should be treated as bad form provided the equation is written in terms of  $x$  at •<sup>2</sup> stage. Otherwise, •<sup>1</sup> is not available.
2. Do not penalise the omission of degree signs.
3. ‘= 0’ must appear by •<sup>3</sup> stage for •<sup>2</sup> to be awarded. However, for candidates using the quadratic formula to solve the equation, ‘= 0’ must appear at •<sup>2</sup> stage for •<sup>2</sup> to be awarded.
4. Candidates may express the equation obtained at •<sup>2</sup> in the form  $6S^2 + S - 1 = 0$ ,  $6x^2 + x - 1 = 0$  or using any other dummy variable at the •<sup>3</sup> stage. In these cases, award •<sup>3</sup> for  $(3S - 1)(2S + 1)$  or  $(3x - 1)(2x + 1)$ .  
However, •<sup>4</sup> is only available if  $\sin x^\circ$  appears explicitly at this stage - see Candidate A.
5. The equation  $1 - 6\sin^2 x^\circ - \sin x^\circ = 0$  does not gain •<sup>2</sup> unless •<sup>3</sup> has been awarded.
6. •<sup>3</sup> is awarded for identifying the factors of the quadratic obtained at •<sup>2</sup> eg “ $3\sin x^\circ - 1 = 0$  and  $2\sin x^\circ + 1 = 0$ ”.
7. •<sup>4</sup> and •<sup>5</sup> are only available as a consequence of trying to solve a quadratic equation - see Candidate B.
8. •<sup>3</sup>, •<sup>4</sup> and •<sup>5</sup> are not available for any attempt to solve a quadratic equation written in the form  $ax^2 + bx = c$  - see Candidate C.
9. •<sup>5</sup> is only available where at least one of the equations at •<sup>4</sup> leads to two solutions for  $x$ .
10. Do not penalise additional (correct) solutions at •<sup>5</sup>. However see Candidates E and F.
11. Accept answers which round to 19, 19.5 and 161.

| Question                                                                                                                                                                                                                                                                                                                  | Generic scheme                                                                                                                             | Illustrative scheme                                                                                                                                                                                                                                                                                                                                             | Max mark                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7. (continued)                                                                                                                                                                                                                                                                                                            |                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                    |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                       |                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                    |
| <b>Candidate A</b><br>$\vdots$<br>$6S^2 + S - 1 = 0$<br>$(3S - 1)(2S + 1) = 0$<br>$S = \frac{1}{3}, S = -\frac{1}{2}$<br>$x = 19.5, 160.5, 210, 330$                                                                                                                                                                      | $\bullet^1 \checkmark \bullet^2 \checkmark$<br>$\bullet^3 \checkmark$<br>$\bullet^4 \wedge$<br>$\bullet^5 \boxed{\checkmark_1}$            | <b>Candidate B - not solving a quadratic</b><br>$\vdots$<br>$6 \sin^2 x^\circ + \sin x^\circ - 1 = 0$<br>$7 \sin x^\circ - 1 = 0$<br>$\sin x^\circ = \frac{1}{7}$<br>$x = 8.2$                                                                                                                                                                                  | $\bullet^1 \checkmark$<br>$\bullet^2 \checkmark$<br>$\bullet^3 \times$<br>$\bullet^4 \boxed{\checkmark_2}$<br>$\bullet^5 \boxed{\checkmark_2}$                     |
| <b>Candidate C - not in standard quadratic form</b><br>$\sin x^\circ + 2 = 3 - 6 \sin^2 x^\circ$<br>$6 \sin^2 x^\circ + \sin x^\circ = 1$<br>$\sin x^\circ (6 \sin x^\circ + 1) = 1$<br>$\sin x^\circ = 1 \quad 6 \sin x^\circ + 5 = 1$<br>$\Rightarrow \sin x = -\frac{4}{6}$<br>$90, 221.8, 318.2$                      | $\bullet^1 \checkmark$<br>$\bullet^2 \boxed{\checkmark_2}$<br>$\bullet^3 \boxed{\checkmark_2}$<br>$\bullet^4 \times$<br>$\bullet^5 \times$ | <b>Candidate D - reading <math>\cos 2x^\circ</math> as <math>\cos^2 x^\circ</math></b><br>$\sin x^\circ + 2 = 3 \cos^2 x^\circ$<br>$\sin x^\circ + 2 = 3(1 - \sin^2 x^\circ)$<br>$3 \sin^2 x^\circ + \sin x^\circ - 1 = 0$<br>$\sin x^\circ = \frac{-1 \pm \sqrt{13}}{6}$<br>$\sin x^\circ = 0.434..., \sin x^\circ = -0.767...$<br>$25.7, 154.3, 230.1, 309.9$ | $\bullet^1 \times$<br>$\bullet^2 \boxed{\checkmark_1}$<br>$\bullet^3 \boxed{\checkmark_1}$<br>$\bullet^4 \boxed{\checkmark_1}$<br>$\bullet^5 \boxed{\checkmark_1}$ |
| <b>Candidate E</b><br>$\vdots$<br>$(3 \sin x^\circ - 1)(2 \sin x^\circ + 1) = 0$<br>$\sin x^\circ = \frac{1}{3}, \sin x^\circ = -\frac{1}{2}$<br>$x = 19, x = 161 \quad x = 30$<br>$x = 210, x = 330$<br><b>However, where the final solution(s) are clearly identified by the candidate award <math>\bullet^5</math></b> | $\bullet^1 \checkmark \bullet^2 \checkmark$<br>$\bullet^3 \checkmark$<br>$\bullet^4 \checkmark$<br>$\bullet^5 \times$                      | <b>Candidate F</b><br>$\vdots$<br>$(3 \sin x^\circ - 1)(2 \sin x^\circ + 1) = 0$<br>$\sin x^\circ = \frac{1}{3}, \sin x^\circ = -\frac{1}{2}$<br>$x = 19, 161, 30, 210, 330$                                                                                                                                                                                    | $\bullet^1 \checkmark \bullet^2 \checkmark$<br>$\bullet^3 \checkmark$<br>$\bullet^4 \checkmark$<br>$\bullet^5 \times$                                              |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                     | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Max mark |
|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8.       |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> integrate using “upper - lower”</li> <li>•<sup>2</sup> identify limits</li> <li>•<sup>3</sup> integrate</li> <li>•<sup>4</sup> substitute limits</li> <li>•<sup>5</sup> calculate shaded area</li> </ul>                                                                                      | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\int \left( (x^3 - 2x^2 - 4x + 1) - (x - 5) \right) dx</math></li> <li>•<sup>2</sup> <math>\int_{-2}^1 \left( (x^3 - 2x^2 - 4x + 1) - (x - 5) \right) dx</math></li> <li>•<sup>3</sup> <math>\frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x</math></li> <li>•<sup>4</sup> <math>\left( \frac{(1)^4}{4} - \frac{2(1)^3}{3} - \frac{5(1)^2}{2} + 6(1) \right) - \left( \frac{(-2)^4}{4} - \frac{2(-2)^3}{3} - \frac{5(-2)^2}{2} + 6(-2) \right)</math></li> <li>•<sup>5</sup> <math>\frac{63}{4}</math> or <math>15\frac{3}{4}</math></li> </ul>                                                                                                                                             | 5        |
|          |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> know to integrate between appropriate limits for both integrals</li> <li>•<sup>2</sup> integrate both functions</li> <li>•<sup>3</sup> substitute limits into both expressions</li> <li>•<sup>4</sup> evaluate both integrals</li> <li>•<sup>5</sup> evidence of subtracting areas</li> </ul> | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\int_{-2}^1 \dots dx</math> and <math>\int_{-2}^1 \dots dx</math></li> <li>•<sup>2</sup> <math>\frac{x^4}{4} - \frac{2x^3}{3} - \frac{4x^2}{2} + x</math> and <math>\frac{x^2}{2} - 5x</math></li> <li>•<sup>3</sup> <math>\left( \frac{(1)^4}{4} - \frac{2(1)^3}{3} - \frac{4(1)^2}{2} + (1) \right) - \left( \frac{(-2)^4}{4} - \frac{2(-2)^3}{3} - \frac{4(-2)^2}{2} + (-2) \right)</math><br/>and<br/><math>\left( \frac{(1)^2}{2} - 5(1) \right) - \left( \frac{(-2)^2}{2} - 5(-2) \right)</math></li> <li>•<sup>4</sup> <math>-\frac{3}{4}</math> and <math>-\frac{33}{2}</math></li> <li>•<sup>5</sup> <math>-\frac{3}{4} - \left( -\frac{33}{2} \right) = \frac{63}{4}</math></li> </ul> |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Generic scheme | Illustrative scheme | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|----------|
| 8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (continued)    |                     |          |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |                     |          |
| <p>1. Correct answer with no working - award 1/5.</p> <p>2. In Method 1, treat the absence of brackets at <math>\bullet^1</math> stage as bad form only if the correct integral is obtained at <math>\bullet^3</math> - see Candidates A and B.</p> <p>3. Do not penalise lack of 'dx' at <math>\bullet^1</math>.</p> <p>4. Limits and 'dx' must appear by the <math>\bullet^2</math> stage for <math>\bullet^2</math> to be awarded in Method 1 and by the <math>\bullet^1</math> stage for <math>\bullet^1</math> to be awarded in Method 2.</p> <p>5. Where a candidate differentiates one or more terms at <math>\bullet^3</math>, then <math>\bullet^3</math>, <math>\bullet^4</math> and <math>\bullet^5</math> are unavailable.</p> <p>6. Accept unsimplified expressions at <math>\bullet^3</math> e.g. <math>\frac{x^4}{4} - \frac{2x^3}{3} - \frac{4x^2}{2} + x - \frac{x^2}{2} + 5x</math>.</p> <p>7. Do not penalise the inclusion of '+c'.</p> <p>8. Do not penalise the continued appearance of the integral sign after <math>\bullet^2</math></p> <p>9. Candidates who substitute limits without integrating do not gain <math>\bullet^3</math>, <math>\bullet^4</math> or <math>\bullet^5</math>.</p> <p>10. <math>\bullet^5</math> is not available where solutions include statements such as '<math>-\frac{63}{4} = \frac{63}{4}</math> square units' - see Candidate B.</p> <p>11. Where a candidate only integrates <math>x^3 - 2x^2 - 4x + 1</math> or another cubic or quartic expression, only <math>\bullet^3</math> and <math>\bullet^4</math> are available (from Method 1).</p> |                |                     |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Generic scheme | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8. (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Candidate A - bad form corrected</b><br>$\int_{-2}^1 x^3 - 2x^2 - 4x + 1 - x - 5 dx \quad \bullet^2 \checkmark$ $= \frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x \quad \bullet^3 \checkmark \Rightarrow \bullet^1 \checkmark$ <p>Bad form at <math>\bullet^1</math> must be corrected by the integration stage and may also take the form of a missing minus sign</p>                                                                                                                            |                | <b>Candidate B</b><br>$\int_{-2}^1 x^3 - 2x^2 - 4x + 1 - x - 5 dx \quad \bullet^1 \times \bullet^2 \checkmark$ $= \frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} - 4x \quad \bullet^3 \boxed{\checkmark}_1$ $= \left( \frac{(1)^4}{4} - \frac{2(1)^3}{3} - \frac{5(1)^2}{2} - 4(1) \right) - \left( \frac{(-2)^4}{4} - \frac{2(-2)^3}{3} - \frac{5(-2)^2}{2} - 4(-2) \right) \bullet^4 \boxed{\checkmark}_1$ $- \frac{57}{4} \text{ cannot be negative so } = \frac{57}{4} \quad \bullet^5 \times$ <p>However, <math>\int \dots = -\frac{57}{4}</math> so Area = <math>\frac{57}{4} \quad \bullet^5 \boxed{\checkmark}_1</math></p> |          |
| <b>Candidate C - lower – upper</b><br>$\int_{-2}^1 ((x-5) - (x^3 - 2x^2 - 4x + 1)) dx \quad \bullet^2 \checkmark$ $= -\frac{x^4}{4} + \frac{2x^3}{3} + \frac{5x^2}{2} - 6x \quad \bullet^3 \checkmark$ $\left( -\frac{(1)^4}{4} + \frac{2(1)^3}{3} + \frac{5(1)^2}{2} - 6(1) \right) - \left( -\frac{(-2)^4}{4} + \frac{2(-2)^3}{3} + \frac{5(-2)^2}{2} - 6(-2) \right) \bullet^4 \checkmark$ $= -\frac{63}{4}$ <p>So Area = <math>\frac{63}{4} \quad \bullet^1 \checkmark \bullet^5 \checkmark</math></p> |                | <b>Candidate D - reversed limits</b><br>$\int_1^{-2} ((x^3 - 2x^2 - 4x + 1) - (x - 5)) dx \quad \bullet^1 \checkmark$ $= \frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x \quad \bullet^3 \checkmark$ $\left( \frac{(-2)^4}{4} - \frac{2(-2)^3}{3} - \frac{5(-2)^2}{2} + 6(-2) \right) - \left( \frac{(1)^4}{4} - \frac{2(1)^3}{3} - \frac{5(1)^2}{2} + 6(1) \right) \bullet^4 \checkmark$ $= -\frac{63}{4}$ <p>So Area = <math>\frac{63}{4} \quad \bullet^2 \checkmark \bullet^5 \checkmark</math></p>                                                                                                                          |          |
| <b>Candidate E - ‘upper’ – ‘lower’</b><br>$= x^3 - 2x^2 - 5x + 6$ $\int_{-2}^1 (x^3 - 2x^2 - 5x + 6) dx \quad \bullet^1 \checkmark \bullet^2 \checkmark$ $= \frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x \quad \bullet^3 \checkmark$ $= \frac{37}{12} - \left( -\frac{38}{3} \right) \bullet^4 \checkmark$ $= \frac{63}{4} \bullet^5 \checkmark$                                                                                                                                                   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                          | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                   | Max mark |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.       | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use compound angle formula</li> <li>•<sup>2</sup> compare coefficients</li> <li>•<sup>3</sup> process for <math>k</math></li> <li>•<sup>4</sup> process for <math>a</math> and express in required form</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>k \sin x^\circ \cos a^\circ + k \cos x^\circ \sin a^\circ</math><br/><b>stated explicitly</b></li> <li>•<sup>2</sup> <math>k \cos a^\circ = -3, k \sin a^\circ = 7</math><br/><b>stated explicitly</b></li> <li>•<sup>3</sup> <math>\sqrt{58}</math></li> <li>•<sup>4</sup> <math>\sqrt{58} \sin(x + 113.19\dots)^\circ</math>.</li> </ul> | 4        |

**Notes:**

1. Do not penalise the omission of degree symbols in this question.
2. Accept  $k(\sin x^\circ \cos a^\circ + \cos x^\circ \sin a^\circ)$  at •<sup>1</sup>.
3. Treat  $k \sin x^\circ \cos a^\circ + \cos x^\circ \sin a^\circ$  as bad form only if the equations at the •<sup>2</sup> stage both contain  $k$ .
4.  $\sqrt{58} \sin x^\circ \cos a^\circ + \sqrt{58} \cos x^\circ \sin a^\circ$  or  $\sqrt{58}(\sin x^\circ \cos a^\circ + \cos x^\circ \sin a^\circ)$  are acceptable for •<sup>1</sup> and •<sup>3</sup>.
5. •<sup>2</sup> is not available for  $k \cos x^\circ = -3$  and  $k \sin x^\circ = 7$ , however •<sup>4</sup> may still be gained - see Candidate E.
6. •<sup>3</sup> is only available for a single value of  $k, k > 0$ .
7. •<sup>4</sup> is not available for a value of  $a$  given in radians.
8. Accept values of  $a$  which round to 113.
9. Candidates may use any form of the wave function for •<sup>1</sup>, •<sup>2</sup> and •<sup>3</sup>. However, •<sup>4</sup> is only available if the wave is interpreted in the form  $k \sin(x + a)^\circ$ .
10. Evidence for •<sup>4</sup> may appear in part (b).

| Question                                                                                                                                                                                                                                                                                                                                       | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                | Illustrative scheme                                                                                                                                                                                                                                                                                                                          | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9. (continued)                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Candidate A</b><br>$\bullet^1 \wedge$<br>$\sqrt{58} \cos a^\circ = -3$<br>$\sqrt{58} \sin a^\circ = 7$ $\bullet^2 \checkmark \bullet^3 \checkmark$<br>$\tan a^\circ = -\frac{7}{3}$<br>$a = 113.19\dots$<br>$\sqrt{58} \sin(x + 113.19\dots)^\circ$ $\bullet^4 \checkmark$                                                                  | <b>Candidate B</b><br>$k \sin x^\circ \cos a^\circ + k \cos x^\circ \sin a^\circ$ $\bullet^1 \checkmark$<br>$\cos a^\circ = -3$<br>$\sin a^\circ = 7$ $\bullet^2$<br>$\times$<br>$\tan a^\circ = -\frac{7}{3}$<br>$a = 113.19\dots$<br>$\sqrt{58} \sin(x + 113.19\dots)^\circ$ $\bullet^3 \checkmark \bullet^4 \times$<br><div>Not consistent with equations at <math>\bullet^2</math>.</div> | <b>Candidate C</b><br>$\sin x^\circ \cos a^\circ + \cos x^\circ \sin a^\circ$ $\bullet^1 \times$<br>$\cos a^\circ = -3$<br>$\sin a^\circ = 7$ $\bullet^2 \boxed{\checkmark}_2$<br>$k = \sqrt{58}$ $\bullet^3 \checkmark$<br>$\tan a^\circ = -\frac{7}{3}$<br>$a = 113.19\dots$<br>$\sqrt{58} \sin(x + 113.19\dots)^\circ$ $\bullet^4 \times$ |          |
| <b>Candidate D - errors at <math>\bullet^2</math></b><br>$k \sin x \cos a + k \cos x \sin a$ $\bullet^1 \checkmark$<br>$k \cos a^\circ = 7$<br>$k \sin a^\circ = -3$ $\bullet^2 \times$<br>$\tan a^\circ = -\frac{3}{7}$<br>$a = 336.80\dots$<br>$\sqrt{58} \sin(x + 336.80\dots)^\circ$ $\bullet^3 \checkmark \bullet^4 \boxed{\checkmark}_1$ | <b>Candidate E - use of <math>x</math> at <math>\bullet^2</math></b><br>$k \sin x \cos a + k \cos x \sin a$ $\bullet^1 \checkmark$<br>$k \cos x^\circ = -3$<br>$k \sin x^\circ = 7$ $\bullet^2 \times$<br>$\tan a^\circ = -\frac{7}{3}$<br>$a = 113.19\dots$<br>$\sqrt{58} \sin(x + 113.19\dots)^\circ$ $\bullet^3 \checkmark \bullet^4 \boxed{\checkmark}_1$                                 | <b>Candidate F</b><br>$k \sin A \cos B + k \cos A \sin B$ $\bullet^1 \times$<br>$k \cos A = -3$<br>$k \sin A = 7$ $\bullet^2 \times$<br>$\tan A = -\frac{7}{3}$<br>$A = 113.19\dots$<br>$\sqrt{58} \sin(x + 113.19\dots)^\circ$ $\bullet^3 \checkmark \bullet^4 \boxed{\checkmark}_1$                                                        |          |

| Question                                                                                                                                                                            |     |      | Generic scheme                                                                                                                                                                                                              | Illustrative scheme                                                                                                                                                                                                                                                                                | Max mark                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 9.                                                                                                                                                                                  | (b) | (i)  | • <sup>5</sup> state maximum value                                                                                                                                                                                          | • <sup>5</sup> $2\sqrt{58}$                                                                                                                                                                                                                                                                        | 1                                                            |
|                                                                                                                                                                                     |     | (ii) | <p><b>Method 1</b></p> <p>•<sup>6</sup> start to solve</p> <p>•<sup>7</sup> state value of <math>x</math></p> <p><b>Method 2</b></p> <p>•<sup>6</sup> start to solve</p> <p>•<sup>7</sup> state value of <math>x</math></p> | <p><b>Method 1</b></p> <p>•<sup>6</sup> <math>x + 113.19... = 90</math><br/>leading to <math>x = -23.19...</math></p> <p>•<sup>7</sup> <math>x = 336.80...</math></p> <p><b>Method 2</b></p> <p>•<sup>6</sup> <math>x + 113.19... = 450</math></p> <p>•<sup>7</sup> <math>x = 336.80...</math></p> | 2                                                            |
| <b>Notes:</b>                                                                                                                                                                       |     |      |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                    |                                                              |
| 11. • <sup>7</sup> is only available where an angle outwith the range $0 \leq x < 360$ needs to be considered - see Candidate G.                                                    |     |      |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                    |                                                              |
| 12. • <sup>7</sup> is only available where • <sup>6</sup> has been awarded. However, see Candidate K.                                                                               |     |      |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                    |                                                              |
| <b>Commonly Observed Responses:</b>                                                                                                                                                 |     |      |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                    |                                                              |
| <b>Candidate G - not considering angle outwith <math>0 \leq x &lt; 360</math></b><br>$\sqrt{58} \sin(x - 23)^\circ$ from part (a)<br>$x - 23 = 90$<br>$x = 113$                     |     |      | • <sup>6</sup> <input checked="" type="checkbox"/> 1<br>• <sup>7</sup> <input checked="" type="checkbox"/> 2                                                                                                                | <b>Candidate H - simplification</b><br>(i) $2\sqrt{58}$<br>(ii) $\sqrt{58} \sin(x + 113)^\circ = \sqrt{58}$<br>$x + 113 = 90$<br>$x = -23$<br>$x = 337$                                                                                                                                            | • <sup>5</sup> ✓<br><br>• <sup>6</sup> ✓<br>• <sup>7</sup> ✓ |
| <b>Candidate I - follow-through marking</b><br>(i) $\sqrt{58}$<br>(ii) $2\sqrt{58} \sin(x + 113)^\circ = \sqrt{58}$<br>$x + 113 = 30$<br>$x = -83$<br>$x = 277$                     |     |      | • <sup>5</sup> ✗<br><br>• <sup>6</sup> <input checked="" type="checkbox"/> 1<br>• <sup>7</sup> <input checked="" type="checkbox"/> 1                                                                                        | <b>Candidate J - graphical approach</b><br>(i) $\sqrt{58}$<br>(ii) max occurs when $x + 113 = 90$<br>$x = -23$<br>$x = 337$                                                                                                                                                                        | • <sup>5</sup> ✗<br><br>• <sup>6</sup> ✓<br>• <sup>7</sup> ✓ |
| <b>Candidate K - no acknowledgement of <math>\times 2</math></b><br>(i) $\sqrt{58}$<br>(ii) $\sqrt{58} \sin(x + 113)^\circ = \sqrt{58}$<br>$x + 113 = 90$<br>$x = -23$<br>$x = 337$ |     |      | • <sup>5</sup> ✗<br><br>• <sup>6</sup> ✗<br>• <sup>7</sup> <input checked="" type="checkbox"/> 1                                                                                                                            |                                                                                                                                                                                                                                                                                                    |                                                              |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |      | Generic scheme                                                                                                                                                                                                                                                                                                          | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Max mark |         |      |         |     |         |         |     |     |     |     |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|------|---------|-----|---------|---------|-----|-----|-----|-----|-----|
| 10.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |      | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> differentiate one term</li> <li>•<sup>2</sup> complete differentiation and interpret condition</li> <li>•<sup>3</sup> determine zeros of quadratic expression</li> <li>•<sup>4</sup> state range with justification</li> </ul>                     | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>6x^2 \dots</math> or <math>+18x \dots</math> or <math>-24</math></li> <li>•<sup>2</sup> <math>6x^2 + 18x - 24 &lt; 0</math></li> <li>•<sup>3</sup> 1 and <math>-4</math></li> <li>•<sup>4</sup> <math>-4 &lt; x &lt; 1</math> with eg labelled sketch</li> </ul>                                                                                                                                                                                                                                                                                                                                                                           | 4        |         |      |         |     |         |         |     |     |     |     |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |      | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> differentiate one term</li> <li>•<sup>2</sup> complete differentiation and determine zeros of quadratic expression</li> <li>•<sup>3</sup> construct nature table(s)</li> <li>•<sup>4</sup> interpret sign of derivative and state range</li> </ul> | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>6x^2 \dots</math> or <math>+18x \dots</math> or <math>-24</math></li> <li>•<sup>2</sup> <math>6x^2 + 18x - 24</math> and 1 and <math>-4</math></li> <li>•<sup>3</sup> <table border="1"> <tr> <td><math>x</math></td><td><math>\dots</math></td><td><math>-4</math></td><td><math>\dots</math></td><td><math>1</math></td><td><math>\dots</math></td></tr> <tr> <td><math>f'(x)</math></td><td><math>+</math></td><td><math>0</math></td><td><math>-</math></td><td><math>0</math></td><td><math>+</math></td></tr> </table> </li> <li>•<sup>4</sup> decreasing when <math>f'(x) &lt; 0</math> so <math>-4 &lt; x &lt; 1</math></li> </ul> | $x$      | $\dots$ | $-4$ | $\dots$ | $1$ | $\dots$ | $f'(x)$ | $+$ | $0$ | $-$ | $0$ | $+$ |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\dots$ | $-4$ | $\dots$                                                                                                                                                                                                                                                                                                                 | $1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\dots$  |         |      |         |     |         |         |     |     |     |     |     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $+$     | $0$  | $-$                                                                                                                                                                                                                                                                                                                     | $0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $+$      |         |      |         |     |         |         |     |     |     |     |     |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |      |                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |         |      |         |     |         |         |     |     |     |     |     |
| 1. At • <sup>3</sup> do not penalise candidates who fail to extract the common factor or who have divided the quadratic inequality by 6.<br>2. • <sup>3</sup> and • <sup>4</sup> are not available to candidates who arrive at a linear expression at • <sup>2</sup> .<br>3. Accept the appearance of $-4$ and $1$ within inequalities for • <sup>3</sup> .<br>4. At • <sup>4</sup> , accept " $x > -4$ , $x < 1$ " together with the required justification. |         |      |                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |         |      |         |     |         |         |     |     |     |     |     |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |         |      |                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |         |      |         |     |         |         |     |     |     |     |     |
| <b>Candidate A</b><br>$6x^2 + 18x - 24 < 0$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$6x^2 + 18x - 24 = 0$<br>$x = -4, 1$ • <sup>3</sup> ✓<br>$-4 < x < 1$ with sketch                  • <sup>4</sup> ✓                                                                                                                                                                                                                                                          |         |      | <b>Candidate B - no initial inequation</b><br>$6x^2 + 18x - 24 = 0$ • <sup>1</sup> ✓ • <sup>2</sup> ✗<br>$x = -4, 1$ • <sup>3</sup> ✓<br>$-4 < x < 1$ with sketch                  • <sup>4</sup> ✗                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |         |      |         |     |         |         |     |     |     |     |     |
| <b>Candidate C</b><br>Decreasing when $f'(x) < 0$<br>$f'(x) = 6x^2 + 18x - 24$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>:                                                                                                                                                                                                                                                                                                                                         |         |      | <b>Candidate D - condition applied after simplification</b><br>$f'(x) = 6x^2 + 18x - 24$ • <sup>1</sup> ✓<br>$x^2 + 3x - 4 < 0$ • <sup>2</sup> ^<br>$x = -4, 1$ • <sup>3</sup> ✓<br>$-4 < x < 1$ with sketch                  • <sup>4</sup> ✓                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |         |      |         |     |         |         |     |     |     |     |     |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | Generic scheme                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                                                                                                                                                   | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> state centre of <math>C_1</math></li> <li>•<sup>2</sup> state centre of <math>C_2</math></li> <li>•<sup>3</sup> calculate distance between centres</li> </ul>     | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(4, -2)</math></li> <li>•<sup>2</sup> <math>(-1, 3)</math></li> <li>•<sup>3</sup> <math>\sqrt{50}</math> or <math>5\sqrt{2}</math> or 7.07...</li> </ul>                                                                                                   | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                       |          |
| 1. Accept $x = 4, y = -2$ for • <sup>1</sup> and $x = -1, y = 3$ • <sup>2</sup> . Do not accept $g = 1, f = -3$ for • <sup>2</sup> .<br>2. Do not penalise lack of brackets in • <sup>1</sup> and • <sup>2</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                       |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                       |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> state radius of <math>C_1</math></li> <li>•<sup>5</sup> calculate radius of <math>C_2</math></li> <li>•<sup>6</sup> demonstrate and communicate result</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>r_1 = \sqrt{37}</math> or 6.08...</li> <li>•<sup>5</sup> <math>r_2 = \sqrt{17}</math> or 4.12...</li> <li>•<sup>6</sup> <math>10.20... &gt; 7.07... (&gt; 1.95...)</math><br/> <math>\therefore</math> circles intersect at two distinct points</li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                       |          |
| 3. Accept $\sqrt{1^2 + 3^2 + 7} = \sqrt{17}$ or $\sqrt{1^2 + (-3)^2 + 7} = \sqrt{17}$ for • <sup>5</sup> . However, do not accept $\sqrt{(-1)^2 + 3^2 + 7} = \sqrt{17}$ .<br>4. At • <sup>6</sup> comparison must be made using decimals. Do not accept $\sqrt{37} + \sqrt{17} > \sqrt{50}$ without any further working.<br>5. Evidence for • <sup>4</sup> and • <sup>5</sup> may be found in part (a).<br>6. For candidates who use simultaneous equations, award • <sup>4</sup> for substitution of $y = x + 1$ into the equation of one of the circles, • <sup>5</sup> for rearranging in standard quadratic form and • <sup>6</sup> for obtaining distinct $x$ -coordinates.<br>7. Do not penalise the omission of “at two distinct points” at • <sup>6</sup> . |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                       |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                       |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                                                                    | Max mark |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 12.      |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> integrate one term</li> <li>•<sup>2</sup> complete integration</li> <li>•<sup>3</sup> substitute for <math>x</math> and <math>y</math></li> <li>•<sup>4</sup> state expression for <math>y</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> eg <math>\frac{8x^4}{4} \dots</math></li> <li>•<sup>2</sup> eg <math>\dots + 3x + c</math></li> <li>•<sup>3</sup> <math>3 = \frac{8 \times (-1)^4}{4} + 3 \times (-1) + c</math></li> <li>•<sup>4</sup> <math>y = 2x^4 + 3x + 4</math></li> </ul> | 4        |

**Notes:**

1. For candidates who omit  $+c$  only •<sup>1</sup> is available.
2. For candidates who differentiate either term, •<sup>2</sup>, •<sup>3</sup>, and •<sup>4</sup> are not available.
3. Do not penalise the appearance of an integral sign and/or  $dx$  at •<sup>2</sup> and •<sup>3</sup>.

**Commonly Observed Responses:**

|                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate A - incomplete substitution</b><br>$y = 2x^4 + 3x + c$ • <sup>1</sup> ✓   • <sup>2</sup> ✓<br>$y = 2(-1)^4 + 3(-1) + c$<br>$c = 4$ • <sup>3</sup> ^<br>$y = 2x^4 + 3x + 4$ • <sup>4</sup> <span style="border: 1px solid red; padding: 0 2px;">✓<sub>1</sub></span> | <b>Candidate B - partial integration</b><br>$y = 2x^4 + 3 + c$ • <sup>1</sup> ✓   • <sup>2</sup> ✗<br>$3 = 2(-1)^4 + 3 + c$ • <sup>3</sup> <span style="border: 1px solid red; padding: 0 2px;">✓<sub>1</sub></span><br>$c = -2$<br>$y = 2x^4 + 1$ • <sup>4</sup> <span style="border: 1px solid red; padding: 0 2px;">✓<sub>1</sub></span> |
| <b>Candidate C - integrating over two lines</b><br>$y = 2x^4 + 3x$ • <sup>1</sup> ✓   • <sup>2</sup> ✗<br>$y = 2x^4 + 3x + c$<br>$3 = 2(-1)^4 + 3(-1) + c$ • <sup>3</sup> ✓<br>$y = 2x^4 + 3x + 4$ • <sup>4</sup> ✓                                                              |                                                                                                                                                                                                                                                                                                                                             |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  | Generic scheme                                                                                                         | Illustrative scheme                                                                                                                                  | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 13.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (a) |  | • <sup>1</sup> calculate concentration                                                                                 | • <sup>1</sup> 9.38... (mg/l)                                                                                                                        | 1        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  |                                                                                                                        |                                                                                                                                                      |          |
| 1. Accept any answer which rounds to 9.4 for • <sup>1</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  |                                                                                                                        |                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  |                                                                                                                        |                                                                                                                                                      |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (b) |  | • <sup>2</sup> substitute<br><br>• <sup>3</sup> write in logarithmic form<br><br>• <sup>4</sup> process for $t$        | • <sup>2</sup> $0.66 = 11 \times e^{-0.0053 t}$<br><br>• <sup>3</sup> $\log_e \frac{0.66}{11} = -0.0053 t$<br><br>• <sup>4</sup> 530.83... (minutes) | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  |                                                                                                                        |                                                                                                                                                      |          |
| 2. Where values other than 0.66 are used in the substitution, • <sup>3</sup> and • <sup>4</sup> are still available.<br>3. Evidence for • <sup>3</sup> must be stated explicitly.<br>4. At • <sup>3</sup> all exponentials must be processed.<br>5. Any base may be used at • <sup>3</sup> stage - see Candidate A.<br>6. Accept $\ln 0.06 = -0.0053 t \ln e$ for • <sup>3</sup> .<br>7. Accept any answer where $530 \leq t \leq 532$ at • <sup>4</sup> .<br>8. • <sup>4</sup> is unavailable where a candidate rounds the value of $\ln 0.06$ to fewer than 2 decimal places.<br>9. The calculation at • <sup>4</sup> must follow from the valid use of exponentials and logarithms at • <sup>2</sup> and • <sup>3</sup> .<br>10. For candidates with no working or who take an iterative approach to arrive at $t = 532$ , $t = 531$ or $t = 530$ award 1/3. However, if, in any iterations $C_t$ is evaluated for $t = 530$ and $t = 531$ leading to a final answer of $t = 531$ (minutes) then award 3/3. |     |  |                                                                                                                        |                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  |                                                                                                                        |                                                                                                                                                      |          |
| <b>Candidate A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | <b>Candidate B</b>                                                                                                     |                                                                                                                                                      |          |
| $0.66 = 11e^{-0.0053t}$ • <sup>2</sup> ✓<br>$0.06 = e^{-0.0053t}$<br>$\log_{10} 0.06 = -0.0053t \log_{10} e$ • <sup>3</sup> ✓<br>$t = 531$ minutes      • <sup>4</sup> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  | $0.66 = 11e^{-0.0053t}$ • <sup>2</sup> ✓<br>$t = 531$ minutes      • <sup>3</sup> ^      • <sup>4</sup> ✓ <sub>1</sub> |                                                                                                                                                      |          |

| Question |     |      | Generic scheme                                                                                             | Illustrative scheme                                                                                                                                                                 | Max mark |
|----------|-----|------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 14.      | (a) | (i)  | • <sup>1</sup> express $A$ in terms of $x$ and $h$                                                         | • <sup>1</sup> $(A=)6x^2 + 10xh$                                                                                                                                                    | 1        |
|          |     | (ii) | • <sup>2</sup> express $h$ in terms of $x$<br><br>• <sup>3</sup> substitute for $h$ and demonstrate result | • <sup>2</sup> $h = \frac{7200 - 6x^2}{10x}$<br><br>• <sup>3</sup> $V = 3x \times 2x \times \left( \frac{7200 - 6x^2}{10x} \right)$ leading to<br><br>$V = 4320x - \frac{18}{5}x^3$ | 2        |

**Notes:**

1. Accept unsimplified expressions for •<sup>1</sup>.
2. •<sup>2</sup> is only available where the (simplified) expression for  $A$  contains at least 2 terms.
3. The substitution for  $h$  at •<sup>3</sup> must be clearly shown for •<sup>3</sup> to be awarded.

**Commonly Observed Responses:**

|         |     |    |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |     |    |     |         |   |   |   |       |   |   |   |   |
|---------|-----|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|-----|---------|---|---|---|-------|---|---|---|---|
|         | (b) |    | <ul style="list-style-type: none"><li>•<sup>4</sup> differentiate</li><li>•<sup>5</sup> equate expression for derivative to 0</li><li>•<sup>6</sup> solve for <math>x</math></li><li>•<sup>7</sup> verify nature</li></ul> | <ul style="list-style-type: none"><li>•<sup>4</sup> <math>4320 - \frac{54}{5}x^2</math></li><li>•<sup>5</sup> <math>4320 - \frac{54}{5}x^2 = 0</math></li><li>•<sup>6</sup> 20</li><li>•<sup>7</sup> table of signs for a derivative<table><tr><td><math>x</math></td><td>...</td><td>20</td><td>...</td></tr><tr><td><math>V'(x)</math></td><td>+</td><td>0</td><td>-</td></tr><tr><td>shape</td><td>/</td><td>—</td><td>\</td></tr></table></li></ul> <p>∴ maximum (when <math>x = 20</math>)</p> | $x$ | ... | 20 | ... | $V'(x)$ | + | 0 | - | shape | / | — | \ | 4 |
| $x$     | ... | 20 | ...                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |     |    |     |         |   |   |   |       |   |   |   |   |
| $V'(x)$ | +   | 0  | -                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |     |    |     |         |   |   |   |       |   |   |   |   |
| shape   | /   | —  | \                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |     |    |     |         |   |   |   |       |   |   |   |   |

**Notes:**

4. For any approach which does not use differentiation award 0/4.
5. •<sup>5</sup> can be awarded for  $\frac{54}{5}x^2 = 4320$ .
6. For candidates who integrate any term at the •<sup>4</sup> stage, only •<sup>5</sup> is available on follow through for setting their 'derivative' to 0.
7. Ignore the appearance of  $-20$  at mark •<sup>6</sup>.
8. Where  $-20$  is considered in a nature table (or second derivative), " $x = 20$ " must be clearly identified as leading to the maximum area.
9. •<sup>6</sup> and •<sup>7</sup> are not available to candidates who state that the maximum exists at a negative value of  $x$ .
10. Do not penalise statements such as "max volume is 20" or "max is 20" at •<sup>7</sup>.

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Generic scheme | Illustrative scheme                                                                                                                                                                                                                                                  | Max mark      |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------|---|----------------|---|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|------|---------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----------------|-----|------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|-----|---------|----------------|---|---|----------------|--|--|------------------------------------------------------------------------|--|
| 14. (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| <b>Candidate A - second derivative</b><br>$V''(x) = -\frac{108}{5}x$<br>$V''(20) < 0 \therefore \text{maximum}$<br><div>•<sup>7</sup> ✓</div>                                                                                                                                                                                                                                                                                                                                                                                    |                | <b>Candidate B - beware of multiplying before equating</b><br>$V'(x) = 4320 - \frac{54}{5}x^2$<br><div>✓</div> $V'(x) = 21600 - 54x^2$<br>$21600 - 54x^2 = 0$<br>$x = 20$<br><div>✓<sub>1</sub></div> <div>•<sup>4</sup><br/>•<sup>5</sup> ✗<br/>•<sup>6</sup></div> |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| <b>Candidate C</b><br>Stationary points when $V'(x) = 0$<br>$V'(x) = 4320 - \frac{54}{5}x^2$<br><div>•<sup>4</sup> ✓ •<sup>5</sup> ✓</div>                                                                                                                                                                                                                                                                                                                                                                                       |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| For the table of signs for a derivative, accept:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| <table><tr><td><math>x</math></td><td><math>20^-</math></td><td><math>20</math></td><td><math>20^+</math></td></tr><tr><td><math>V'(x)</math></td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table>                                                                                                                                                                                                                                                                         | $x$            | $20^-$                                                                                                                                                                                                                                                               | $20$          | $20^+$        | $V'(x)$       | +             | 0       | - | Slope or shape |   |   |   | <table><tr><td><math>x</math></td><td><math>\rightarrow</math></td><td><math>20</math></td><td><math>\rightarrow</math></td></tr><tr><td><math>V'(x)</math></td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table> | $x$ | $\rightarrow$ | $20$ | $\rightarrow$ | $V'(x)$ | +                                                                                                                                                                                                                                                                                                                                        | 0   | -   | Slope or shape |     |      |     | <table><tr><td><math>x</math></td><td><math>a</math></td><td><math>20</math></td><td><math>b</math></td></tr><tr><td><math>V'(x)</math></td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table> | $x$ | $a$ | $20$ | $b$ | $V'(x)$ | +              | 0 | - | Slope or shape |  |  |                                                                        |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $20^-$         | $20$                                                                                                                                                                                                                                                                 | $20^+$        |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $V'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | +              | 0                                                                                                                                                                                                                                                                    | -             |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\rightarrow$  | $20$                                                                                                                                                                                                                                                                 | $\rightarrow$ |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $V'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | +              | 0                                                                                                                                                                                                                                                                    | -             |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $a$            | $20$                                                                                                                                                                                                                                                                 | $b$           |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $V'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | +              | 0                                                                                                                                                                                                                                                                    | -             |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| Arrows are taken to mean<br>'in the neighbourhood of'                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                | Where $a < 20$ and $b > 20$                                                                                                                                                                                                                                          |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| For the table of signs for a derivative, accept:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| <table><tr><td><math>x</math></td><td><math>\rightarrow</math></td><td><math>-20</math></td><td><math>\rightarrow</math></td><td><math>20</math></td><td><math>\rightarrow</math></td></tr><tr><td><math>V'(x)</math></td><td>-</td><td>0</td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td><td></td><td></td></tr></table>                                                                                                                                                           | $x$            | $\rightarrow$                                                                                                                                                                                                                                                        | $-20$         | $\rightarrow$ | $20$          | $\rightarrow$ | $V'(x)$ | - | 0              | + | 0 | - | Slope or shape                                                                                                                                                                                                                                                         |     |               |      |               |         | <table><tr><td><math>x</math></td><td><math>a</math></td><td><math>-20</math></td><td><math>b</math></td><td><math>20</math></td><td><math>c</math></td></tr><tr><td><math>V'(x)</math></td><td>-</td><td>0</td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td><td></td><td></td></tr></table> | $x$ | $a$ | $-20$          | $b$ | $20$ | $c$ | $V'(x)$                                                                                                                                                                                                                                            | -   | 0   | +    | 0   | -       | Slope or shape |   |   |                |  |  | Since the function is continuous<br>$-20 \rightarrow 20$ is acceptable |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\rightarrow$  | $-20$                                                                                                                                                                                                                                                                | $\rightarrow$ | $20$          | $\rightarrow$ |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $V'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | -              | 0                                                                                                                                                                                                                                                                    | +             | 0             | -             |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $a$            | $-20$                                                                                                                                                                                                                                                                | $b$           | $20$          | $c$           |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| $V'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | -              | 0                                                                                                                                                                                                                                                                    | +             | 0             | -             |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | Since the function is continuous<br>$-20 < b < 20$ is acceptable                                                                                                                                                                                                     |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |
| <ul style="list-style-type: none"><li>For this question do not penalise the omission of 'x' or the word 'shape'/'slope'.</li><li>Stating values of <math>V'(x)</math> is an acceptable alternative to writing '+' or '-' signs.</li><li>Acceptable variations of <math>V'(x)</math> are: <math>V'</math>, <math>\frac{dV}{dx}</math>, and <math>4320 - \frac{54}{5}x^2</math>. Accept <math>\frac{dy}{dx}</math> only where candidates have previously used <math>y = 4320x - \frac{18}{5}x^3</math> in their working.</li></ul> |                |                                                                                                                                                                                                                                                                      |               |               |               |               |         |   |                |   |   |   |                                                                                                                                                                                                                                                                        |     |               |      |               |         |                                                                                                                                                                                                                                                                                                                                          |     |     |                |     |      |     |                                                                                                                                                                                                                                                    |     |     |      |     |         |                |   |   |                |  |  |                                                                        |  |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  | Generic scheme                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                     | Max mark                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 15.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> determine gradient of tangent</li> <li>•<sup>2</sup> determine gradient of radius</li> <li>•<sup>3</sup> strategy to find centre</li> <li>•<sup>4</sup> state coordinates of centre</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>-\frac{1}{3}</math></li> <li>•<sup>2</sup> 3</li> <li>•<sup>3</sup> eg <math>y = 3x - 1</math> or <math>3 = \frac{y-5}{x-2}</math></li> <li>•<sup>4</sup> (0, -1)</li> </ul> | 4                                                                                    |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                         |                                                                                      |
| 1. Ignore errors in processing the constant term in • <sup>1</sup> .<br>2. Do not accept $m = -\frac{1}{3}x$ for • <sup>1</sup> . However • <sup>2</sup> , • <sup>3</sup> and • <sup>4</sup> are still available where the candidate uses a numerical value for $m_{\perp}$ .<br>3. Accept $y-5 = 3(x-2)$ as evidence for • <sup>3</sup> .<br>4. • <sup>4</sup> is only available as a consequence of trying to find and use a perpendicular gradient along with a point on the $y$ -axis.<br>5. Where candidates use “stepping out” with the perpendicular gradient, the diagram must be consistent with the solution to gain • <sup>3</sup> and • <sup>4</sup> .<br>6. Accept “ $x = 0$ ”, “ $y = -1$ ” stated explicitly for • <sup>4</sup> . |  |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                         |                                                                                      |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                         |                                                                                      |
| <b>Candidate A - perpendicular gradient clearly stated</b><br>$x + 3y = 17$<br><br>$m_{\perp} = 3$<br>$y = 3x - 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  | • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>• <sup>3</sup> ✓                                                                                                                                                                                               | <b>Candidate B - no communication for perpendicular gradient</b><br>$x + 3y = 17$<br>$y = -\frac{1}{3}x + \frac{17}{3}$<br>$m = 3$<br>$y = 3x - 1$                                                                                      | • <sup>1</sup> ^ • <sup>2</sup> ✓<br>• <sup>3</sup> ✓<br>• <sup>4</sup> is available |
| <b>Candidate C - no communication for perpendicular gradient or rearrangement</b><br>$x + 3y = 17$<br>$m = 3$<br>$y = 3x - 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  | • <sup>1</sup> ^ • <sup>2</sup> ^<br>• <sup>3</sup> ✓<br>• <sup>4</sup> is not available                                                                                                                                                            | <b>Candidate D - using geometry</b><br>∴<br>Using point diametrically opposite (2,5), by symmetry identify that $x$ -coordinate is -2.<br>∴ $y = 3(-2) - 1 = -7$ .<br>Centre is midpoint of (-2, -7) and (2,5).<br>∴ centre is (0, -1)  | • <sup>1</sup> ✓ • <sup>2</sup> ✓ • <sup>3</sup> ✓<br>• <sup>4</sup> ✓               |
| <b>Candidate E - incorrect gradient</b><br>$x + 3y = 17$<br>$3y = -x + 17$<br>$m_{\perp} = 1$<br>$1 = \frac{5-y}{2-0}$<br>Centre is at (0,3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  | • <sup>1</sup> ^ • <sup>2</sup> ✗<br>• <sup>3</sup> ✓<br>• <sup>4</sup> ✓                                                                                                                                                                           |                                                                                                                                                                                                                                         |                                                                                      |

[END OF MARKING INSTRUCTIONS]